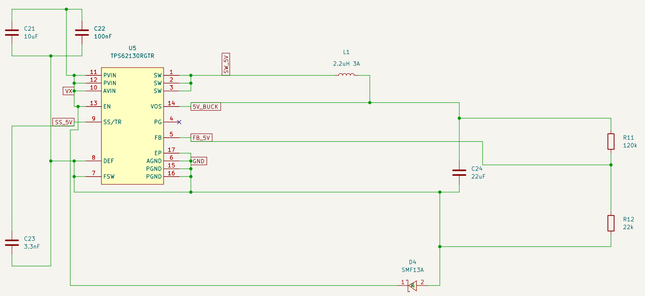
NEXUS-STYLE FLIGHT CONTROLLER | FUNCTIONAL SCHEMATIC

Read left to right: POWER / MCU + SENSORS / INTERFACES. Matching global labels connect across panels.

UNFINISHED PCB CHECKPOINT - NOT RELEASED FOR FABRICATION. X = intentionally unconnected pin.

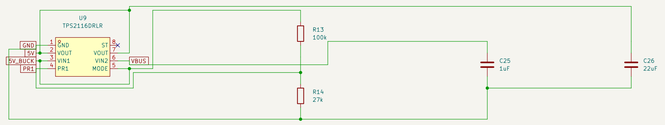
01 VX INPUT / 5 V BUCK

VX -> U5 / L1 -> 5V_BUCK; feedback R11 / R12 sets about 5.16 V



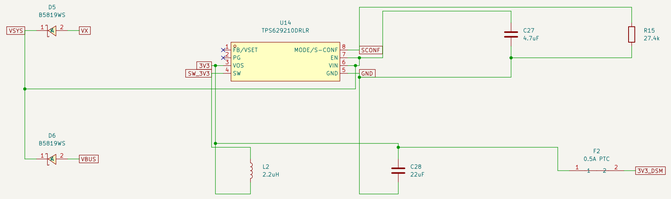
02 5 V SOURCE PRIORITY

5V_BUCK is preferred; USB VBUS supplies 5V when the buck is absent



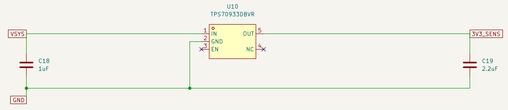
03 VSYS / MCU 3.3 V

VX / VBUS diode-OR -> VSYS -> U14 / L2 -> 3V3; F2 limits the DSM port



04 LOW-NOISE SENSOR SUPPLY

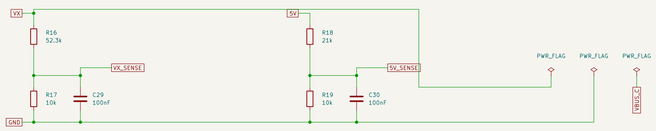
VSYS -> U10 -> 3V3_SENS. EN pin 3 intentionally floating (internal pull-up).



Do not tie EN to VSYS. Pin 4 is also intentionally unconnected.

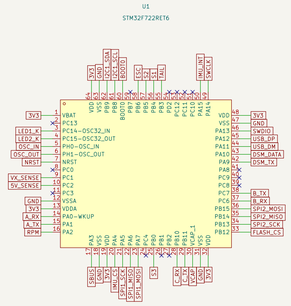
05 VOLTAGE MONITORING / POWER FLAGS

ADC input dividers: VX_SENSE -> U1.9; 5V_SENSE -> U1.10



06 STM32F722 CORE

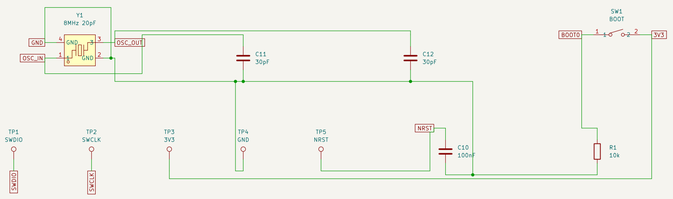
Rotorflight RDMS-NEXUS_F7 pin map; named interfaces continue in panels 09-15



SPI1: IMU | SPI2: blackbox | I2C1: barometer

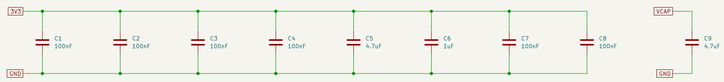
07 CLOCK / BOOT / DEBUG

8 MHz HSE; BOOT selects DFU. SWD pads are for bring-up and debug.



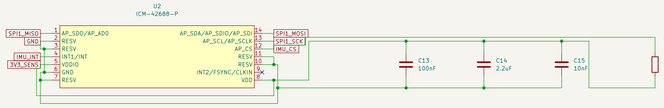
08 MCU DECOUPLING

C1-4: VDD; C5: bulk; C6-7: VDDA; C8: VBAT; C9: VCAP



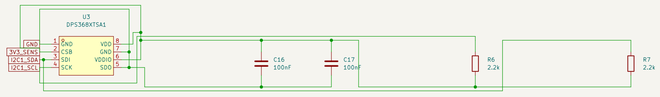
09 SPI1 INERTIAL SENSOR

ICM-42688-P: IMU_CS / SPI1: IMU_INT -> U1.50; powered by 3V3_SENS



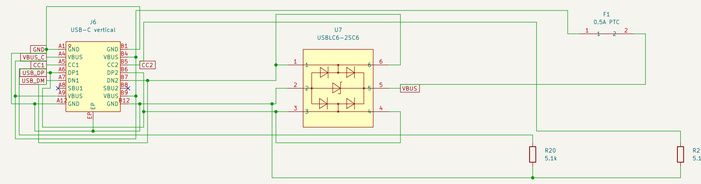
10 I2C1 PRESSURE SENSOR

DPS368: address 0x76; CSB high selects I2C; R6 / R7 pull up SCL / SDA



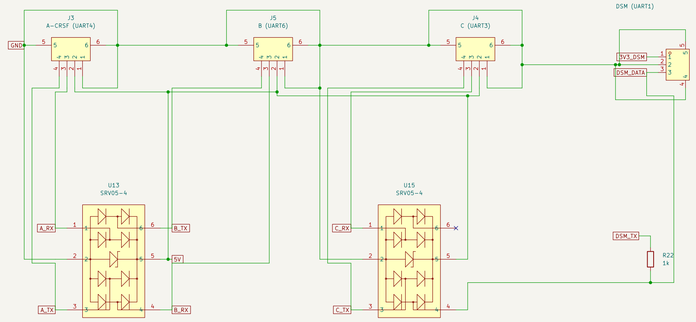
11 USB-C / PROTECTION

USB 2.0 device; CC1 / CC2 Rd = 5.1k; F1 protects VBUS



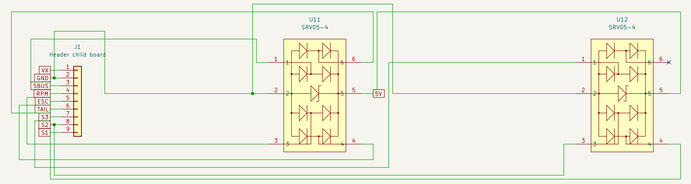
12 RECEIVER INTERFACES

A = UART4 (CRSF); B = UART6; C = UART3; DSM = UART1



13 SERVO HEADER / SIGNAL CLAMPS

J1 mates with the separate header board; VX and GND enter at pins 1 / 2



14 SPI2 BLACKBOX

W25N01GV 128 MB; /CS, /WP and /HOLD pull-ups; EP intentionally floating



15 STATUS INDICATION

PC14 / PC15 sink only: active low, about 1.3 mA. Blue = 5 V bus present.



Rotorflight RDMS-NEXUS_F7 target; LCSC part numbers in the LCSC field
Generated from tools/design.py - edit the design file, then regenerate

Sheet: /
File: nexus-style-fc.kicad_sch

Title: Nexus-style FBL controller - main board

Size: A1 | Date: 2026-10-13 | Rev: 0.2

KiCad E.D.A. 10.0.7 | Id: 1/1